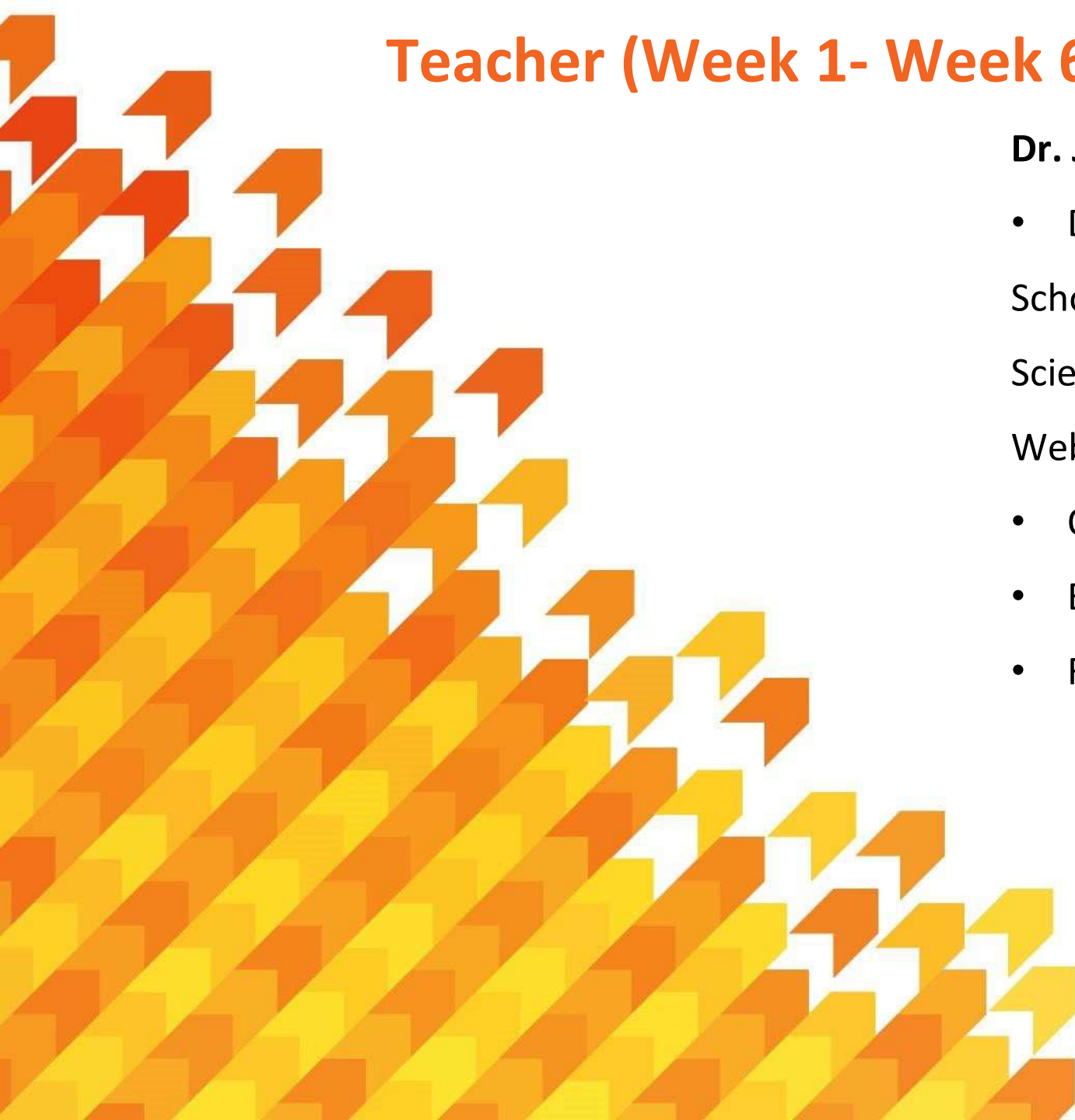


ENME601 Thermodynamics

Week 1 : Introduction and Basics Concept

Presenter: Dr. Jay Wang

2



Teacher (Week 1- Week 6)

Dr. Jay Wang, *Ph.D., M.Phil., M.Eng*

- Department of Mechanical Engineering

School of Engineering, Computer, and Mathematical
Sciences

Webpage: <https://academics.aut.ac.nz/jay.wang>

- Office location: WS-6, WZ 1003 (by appointment)
- Email: jay.wang@aut.ac.nz
- Research areas of interest:
 - Eco-friendly cooling and heating technologies
 - Solar thermal technologies
 - Thermal energy storage
 - Waste heat recovery
 - Power-to-X technology

Course Introduction (1)

- **Delivery of material**

- **In-class lectures/tutorials**

- You will get the most out of this course if you regularly attend lectures and tutorials.*

- Weeks 1 -- 6: Thermodynamics 1 (Dr. Jay Wang).
 - Weeks 7 -- 12: Thermodynamics 2 (Dr. Reza Enayatollahi).

- **Practical Labs: two during semester**

- Saturation pressure (in Week 3) - Check your timeslot and review the lab manual in advance.
 - Rankin Cycle (in Week 8)

Course Introduction (2)

- **Assessments**

- Two practical labs (20%)
- Mid-semester test (30%) -- covering contents from Weeks 1--6.
- Final exam (50%) -- covering contents from Weeks 7--12.

Course Introduction (3)

- **Timetable**

- 2 hour lecture each week (Monday 12 pm – 2 pm) -- WZ 416
- 1 hour tutorial (Thursday, 2 pm – 3 pm) -- WZ 416
- Please check your timetable regularly for any changes (e.g.: lecture/tutorial place)

Course Introduction (4)

- **Canvas**

- Weekly pages including: Lecture slides, recording of lectures, tutorial problems & solutions
- Other resources and information

- **Relevant Textbook**

- *Fundamentals of Thermal-Fluid Sciences*.
Cengel Y A, Cimbala, J M, Ghajar A J. 6th Edition, McGraw Hill, 2022.
- *Thermodynamics: An Engineering Approach*
Yunus A. Çengel, Michael A. Boles, Mehmet Kanoğlu. 9th Edition, McGraw Hill, 2019

- **Communication (Weeks 1 – 6)**

- Course Announcements
- Email: Jay.wang@aut.ac.nz
- Office: WZ1003 (by appointment)
- TA 1: Mr. Cameron Gillingham, Email: dws0135@autuni.ac.nz
- TA 2: Mr. Mahir Ashef Hridoy, Email: ncn9717@autuni.ac.nz

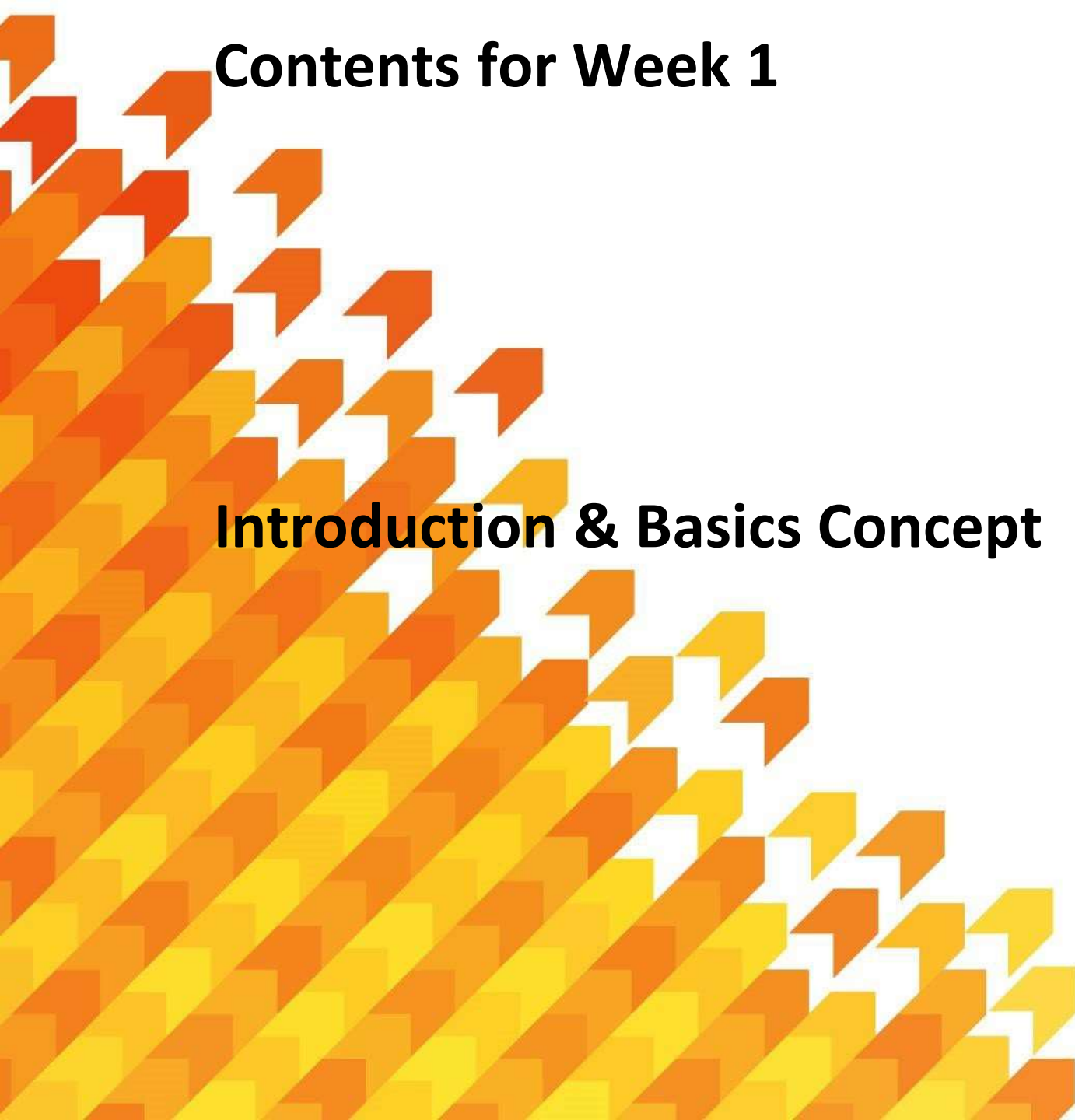
Course Introduction (5)

- **Mid-term test (proposed version)**

- **Individual 2-hour written exam**

This is a **paper-based**, **open-book** exam where all handwritten or printed notes are allowed. Access to the internet and use of electronic devices are not allowed, except for the use of a calculator. All answers must be written in the provided answer booklet.

Draft sheets will be supplied but will not be evaluated.



Contents for Week 1

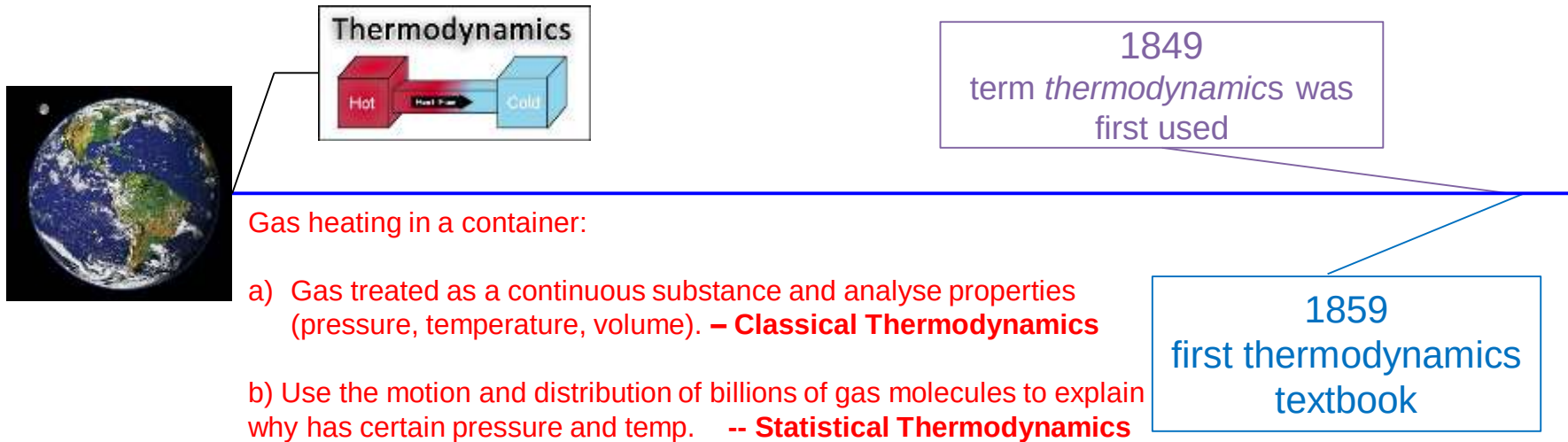
Introduction & Basics Concept

- Thermal and energy
- Application areas of Thermodynamics and Energy
- Dimensions and units
- Short practice 1
- Systems and Control Volumes
- Short practice 2
- Properties of a System
- Processes and Cycles
- Temperature and Pressure

Thermodynamics and energy (1)

Thermodynamics is the science dealing with the relations between all forms of *energy*.

Today, Thermodynamics strongly focuses on energy conversion systems (e.g. refrigerators, power plants).



Classical Thermodynamics: A **macroscopic approach** to the study of Thermodynamics (**it is used in this course**). It involves the concept of continuum.

- Large-scale properties and behaviors



Statistical Thermodynamics: A **microscopic approach** to the study of Thermodynamics.

- Considering the numbers of possible arrangements of atoms within a system, known as microscopic states.

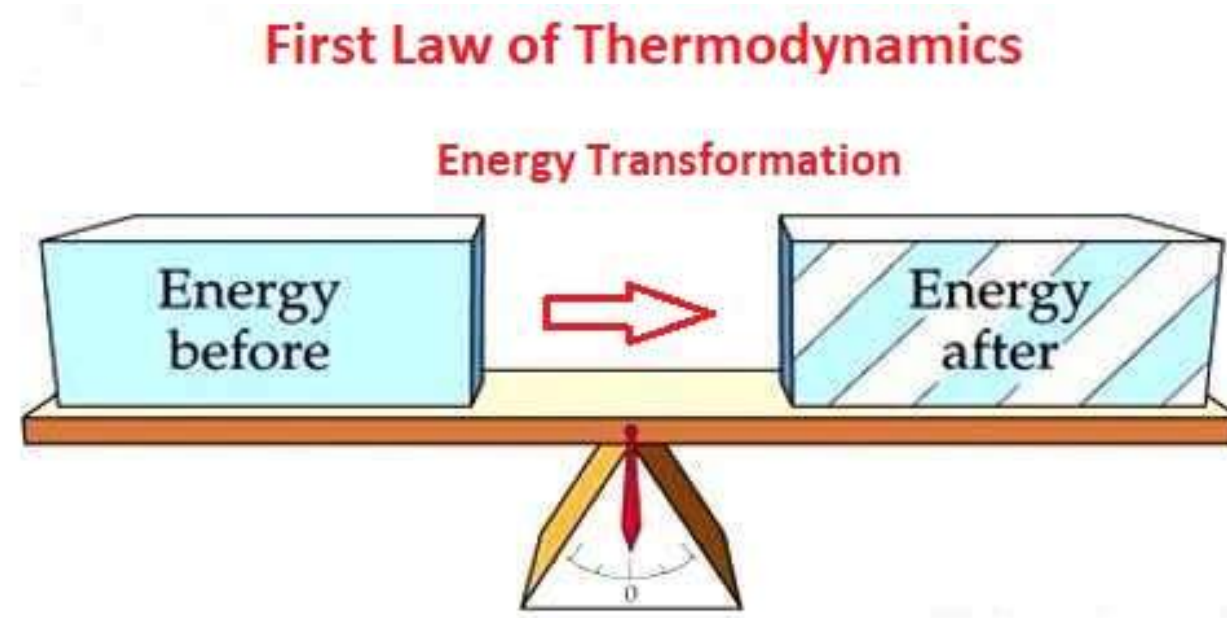
Thermodynamics and energy (2)

- The name **thermodynamics** stems from the Greek words *therme* (heat) and *dynamis* (power).

Q: Thermodynamics Vs. Heat transfer ?

- **Conservation of energy principle:**
 - Energy can change from one form to another, but the total amount of energy remains constant.
 - Energy cannot be created or destroyed.
- **The first law of thermodynamics:**
 - An expression of the conservation of energy principle. - Quantity

Thermodynamics and energy (3)



Energy conservation

Simplified Version: Total **quantity** of energy cannot be created nor destroyed, but can be transferred from one body to another, or **can change from one form to another.**

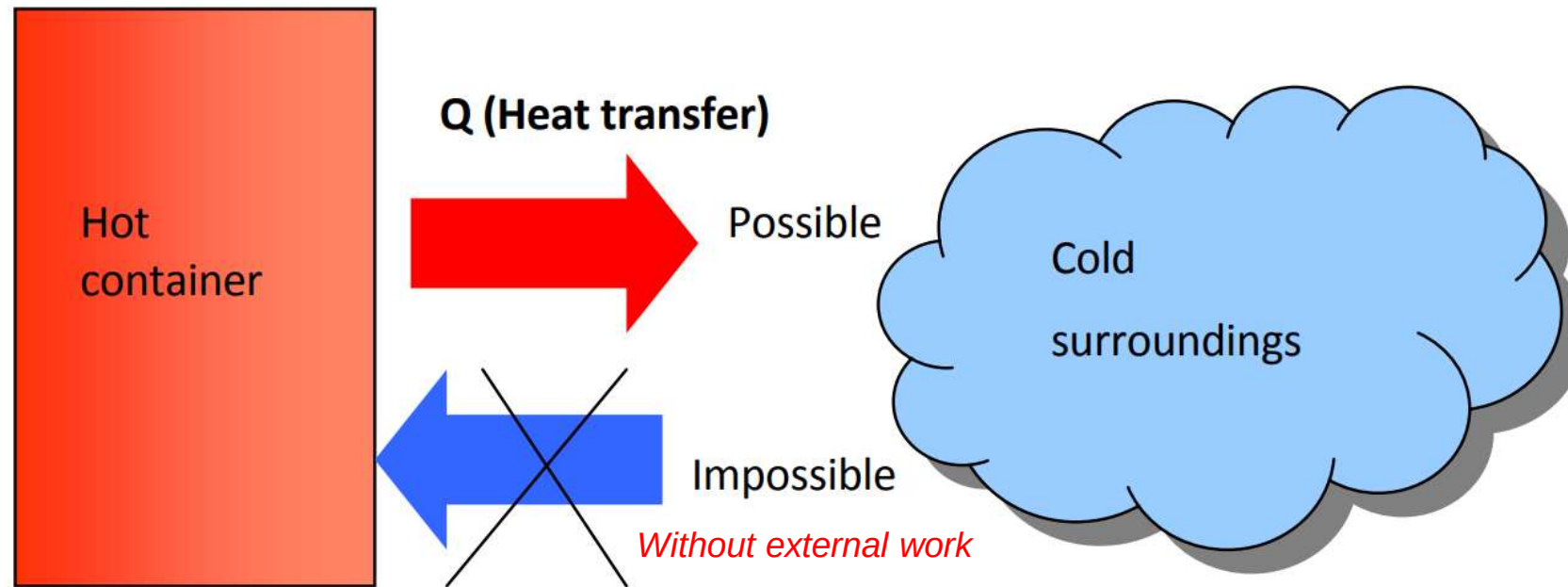
Thermodynamics and energy (4)

- The second law of thermodynamics:

When energy is converted from one form to another or matter moves within a closed system, the system's entropy (disorder or randomness) **increases.**

- It asserts that energy also has *quality*, and actual processes occur in the direction of decreasing quality of energy.

Thermodynamics and energy (5)



2nd law of Thermodynamics:

Quality (able to work) : Heat flows spontaneously from a higher-temperature body to a lower-temperature body. The reverse process cannot occur without external work.

Thermodynamics and energy (6)

A heat process can occur only when:

1st Law of Thermodynamics (quantity) ✓

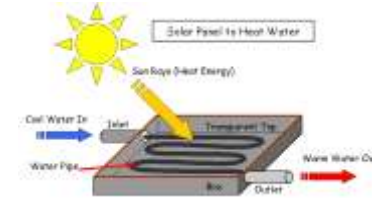
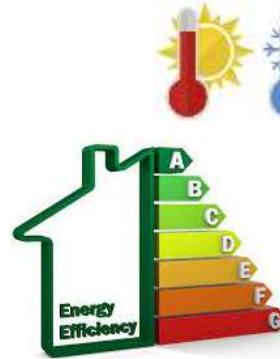
2nd Law of Thermodynamics (quality) ✓

Application areas of Thermodynamics and Energy (1)

Renewable energy; Energy conversion and storage; Intelligent Energy System (System modelling, Automatic control, AI)

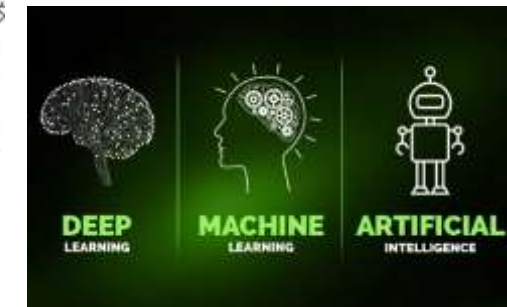
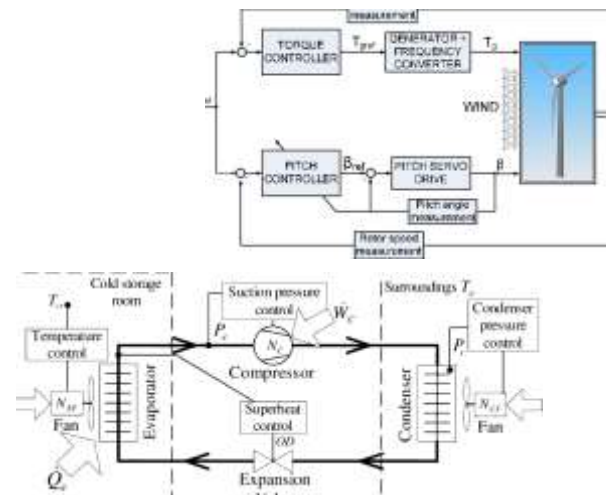


Mechanical engineers play a crucial role in the green energy transition.



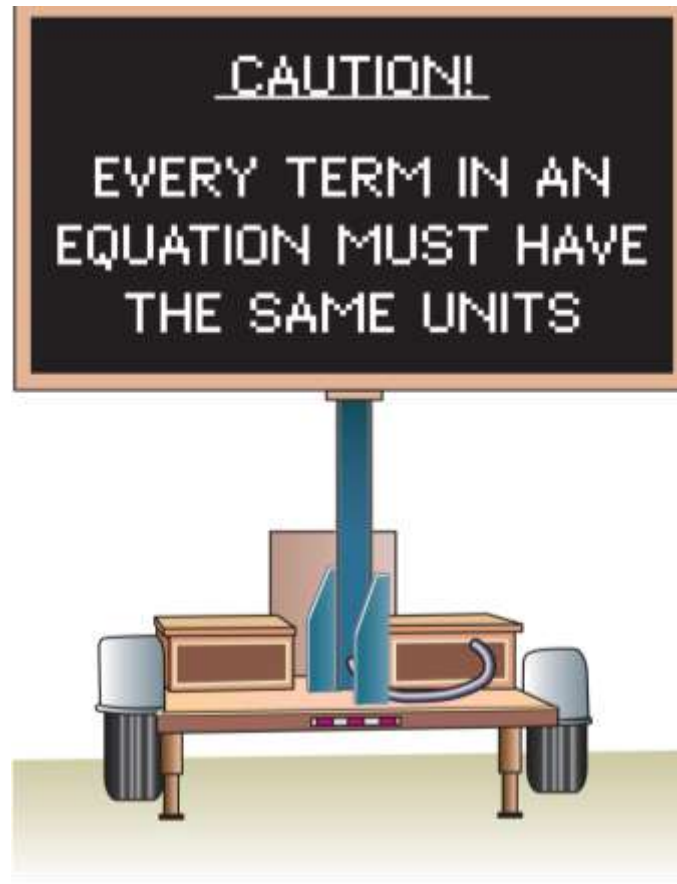
Power plants
© Malcolm Fife/Getty Images RF

...and so do mechatronics engineers!



The importance of dimensions and units (1)

- BE familiar with basic dimensions for mass, length, time, energy etc.
- You MUST use **dimensions and units in solving problems** and ALWAYS state the units for the final answer.
- Use **unit conversion ratios** to determine and ensure the correct units for any answers you state.



Mechanical equivalent of heat ([Video](#))

Short practice (1)

Question: The density of a material can be calculated from this equation: $\rho = m/V$ (kg/m³) where ρ = density, m = mass (kg), V = volume (m³).

To determine the density of a liquid in kg/m³ that has a volume of 900 ml and a mass of 800 g.

Short practice (1) – Solution

Given: Volume (V) = 900 ml, Mass (m) = 800 g, Density formula: $\rho = m / V$

Step 1: Convert units

Mass: 800 g = 800 / 1000 = 0.8 kg

Volume: 900 ml = 900 / 10⁶ = 0.0009 m³

Step 2: Plug into the formula

$\rho = 0.8 \text{ kg} / 0.0009 \text{ m}^3$

$\rho = 888.89 \text{ kg/m}^3$

Final Answer:

$\rho = 888.89 \text{ kg/m}^3$

Problem solving technique (self-reading)

It is really helpful to follow a step by step approach when solving problems

- **Step 1: Problem Statement.**
 - Restate the problem briefly in your own words
 - Note down all the key information provided
 - Note what quantities need to be found (i.e. the 'answers').
- **Step 2: Schematic**
 - Draw a rough sketch of the physical situation
 - Include any relevant information or values on the figure (this could include the key information from Step 1, i.e. combine these two steps).
 - Clearly show the system boundary and indicate any energy and mass interactions across the boundary.
 - If any properties are kept constant (e.g. volume, temperature or pressure), indicate this on your sketch.
- **Step 3: State Assumptions and Approximations**
 - State any assumptions and approximations you make to make the problem solvable. Justify these if necessary. (e.g. ignore kinetic and potential energy changes...)

Problem solving technique (self-reading)

- **Step 4: Physical Laws – relevant equations**
 - **State the equations** (physical laws) relevant to the problem (that allow you to calculate the answers) and simplify by applying the assumptions.
 - Be careful to apply the equations to the region as specified by the system boundary in your sketch.
- **Step 5: Properties**
 - **Determine any unknown properties (at known states)** that are needed to calculate the answers. This may involve using thermodynamic tables or other property relations (equations). Clearly show how you determined these unknown properties.
- **Step 6: Calculations**
 - **Substitute the values** provided in the problem statement (Step 1) and determined unknown properties (Step 5) into the simplified equations (Step 4) and so determine the **answers**.
 - Pay careful attention to the units and **ensure units are consistent** (dimensional homogeneity).
 - Use 5 or so significant figures for calculated input values and final answers should be 3 or 4 significant figures at most (or equivalent to precision of input values).
 - Lay out your solution in as **logical** a manner as possible (easy for someone else to follow).

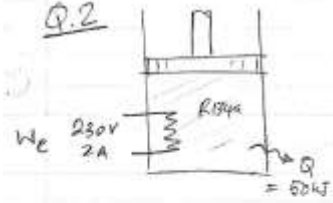
Problem solving technique (self-reading)

- Step 7: Checking, Reasoning, Verification, Discussion
 - **Check** to see that your answers seem reasonable and intuitive.
 - Possibly carry out a rough calculation to see if the answer is of the same order of magnitude.
 - If possible, verify the validity of your assumptions.
 - If necessary, discuss the significance of your results.

This problem solving process will be demonstrated in the examples provided in tutorials and recordings. It is important that you develop your own technique!



Q.2



Initial conditions: $R-134a$
 $V_1 = 0.02 \text{ m}^3$
 $P_1 = 900 \text{ kPa}$ ($T_{sat} = 35.53^\circ\text{C}$)
 $T_1 = 20^\circ\text{C} \Rightarrow$ subcooled

Final conditions:
 $x = 0.5$ (i.e. saturated mix)
 $P_2 = 900 \text{ kPa}$

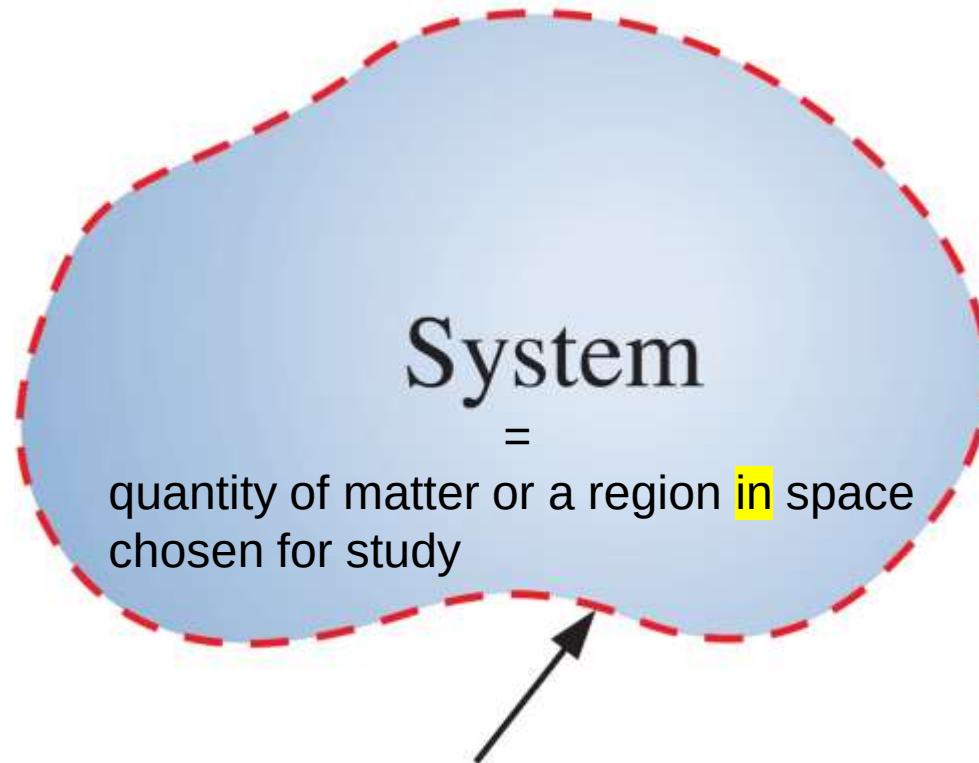
$W_e = V \cdot I \cdot \Delta t$
 $= 230 \times 2 \times \Delta t$

First Law: energy balance: $Q - W = \Delta U$
 $\Rightarrow Q - W_e - W_b = \Delta U$
 $\Rightarrow Q - W_e = \Delta U + W_b$
 $= \Delta H$ (const. P process, $W_b = P\Delta V$)
 $\Rightarrow Q - W_e = \Delta H$
 $\Rightarrow Q - W_e = m(h_2 - h_1)$ (4)

From tables: initial: $h_1 \approx h_f @ 20^\circ\text{C}$ (subcooled liq)
 $= 77.26 \text{ kJ/kg}$ /4

Systems and Control Volumes (1)

Surroundings = mass or region **outside** the system



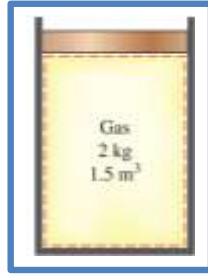
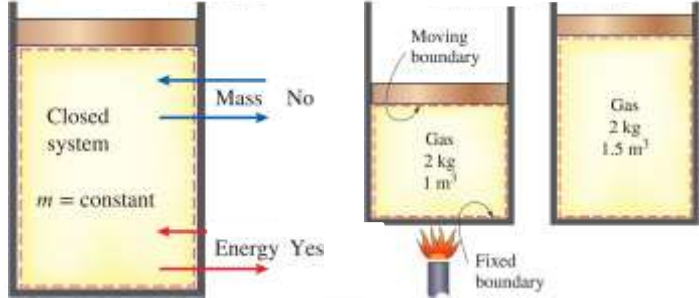
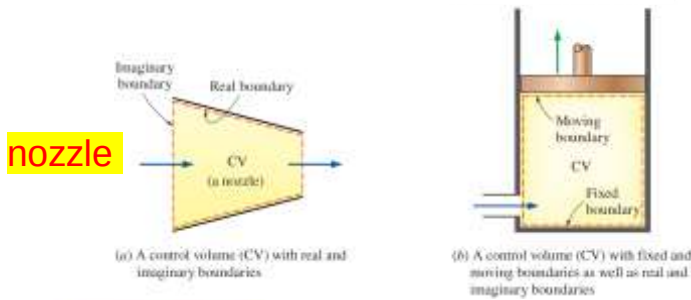
Boundary = real or imaginary surface that **separates** the system from its surroundings and can be *fixed* or *movable*

Closed system (or control mass): A fixed amount of mass and no mass **can** cross its boundary.

Open system (or control volume): A properly selected region in space. Both mass and energy **can** cross the boundary (i.e. **control surface**) of a control volume.

Isolated system: Both mass and energy **cannot** cross the boundary of a system.

Systems and Control Volumes (3)

Isolated	No mass flow No energy flow	
Closed	No mass flow Energy flow in form of heat and work	
Open / Control volume	Energy flow Mass flow	

Mass cannot cross the boundaries of a closed system (Piston & cylinder) but energy can.

1. The cylinder walls form fixed boundaries, while the piston is a moving boundary.

2. As the piston moves, the size of the control volume changes, allowing mass and energy to cross the system boundary.

Short practice (2)

Are these systems isolated, open or closed?

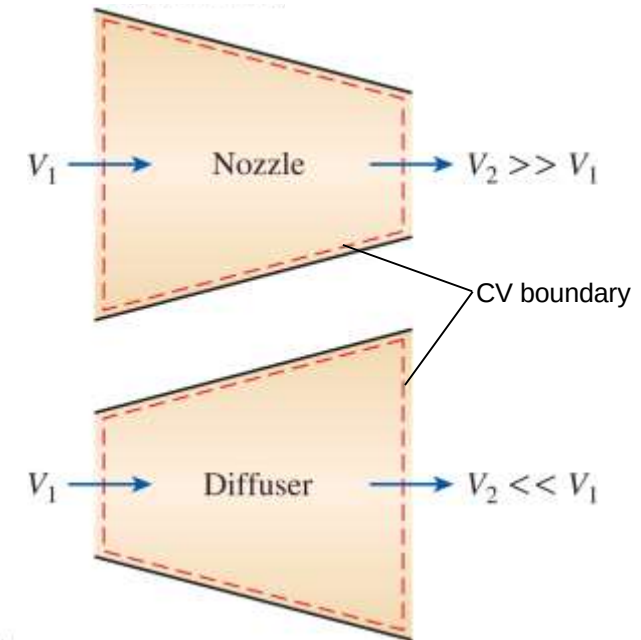
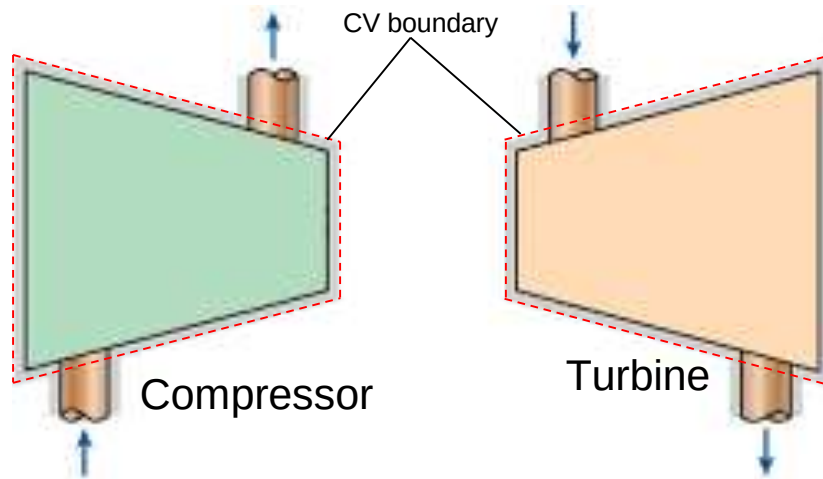


Short practice 2 – Solution

System	Heat Exchange	Mass Exchange	Type
Open Mug	Yes	Yes	Open System
Covered Mug (loose)	Yes	Yes	Open System
Covered Mug (tightly)	Yes	Limited	Closed System
Thermos	No	No	Isolated System

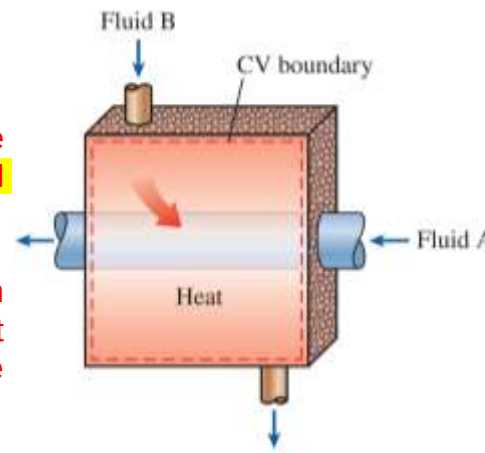
Systems and Control Volumes (4)

Examples of control volumes

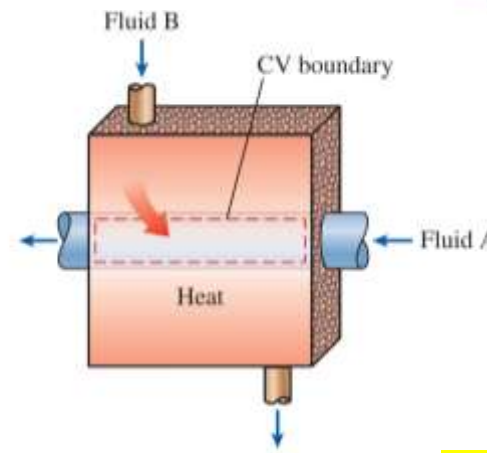


(a) The control volume includes the entire heat exchanger, with both Fluid A and Fluid B inside the system.

Heat is transferred internally between the two fluids, so there is no heat transfer across the control volume boundary ($Q_{CV} = 0$).



(a) System: Entire heat exchanger ($Q_{CV} = 0$)



(b) System: Fluid A ($Q_{CV} \neq 0$)

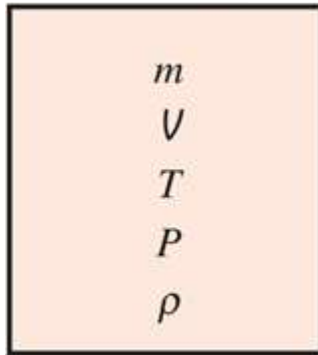
(b) The control volume includes only Fluid A. (Fluid B is outside the CV)

Heat is transferred from Fluid B across the control volume boundary into Fluid A, so heat crosses the boundary ($Q_{CV} \neq 0$).

Key takeaway: The same physical device can have different control volumes depending on the analysis objective.

Properties of a System (1)

A **property** is a macroscopic characteristic of a system, such as mass (m), volume (V), energy (E), pressure (P), density (ρ), and temperature (T).



Extensive properties: Properties whose values **depend on** the size or extent of the system, such as mass (m), volume (V) and energy (E).

Intensive properties: Properties whose values are **independent of the mass of a system**, such as temperature (T), pressure (P) and density (ρ).



- Extensive properties denoted with uppercase letters (excepts for mass m)
- Intensive properties denoted with lowercase letters (except for pressure P and temperature in Kelvin T)

The **specific properties** represent extensive properties per unit mass, such as specific energy ($e = E/m$) and specific volume ($v = V/m$).

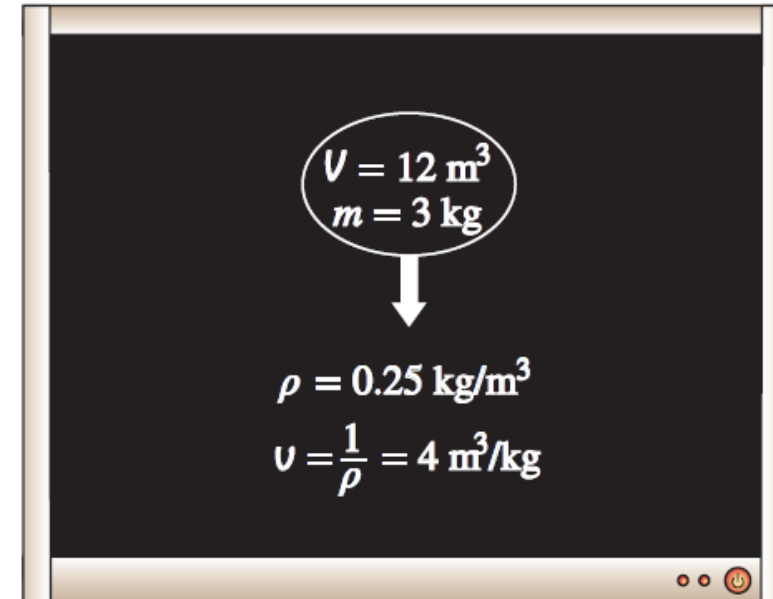
Properties of a System (2)

- **Density (ρ)** is defined as mass per unit volume

$$\rho = \frac{m}{V} \quad (\text{kg/m}^3)$$

- **Specific volume (v)** is the reciprocal of density

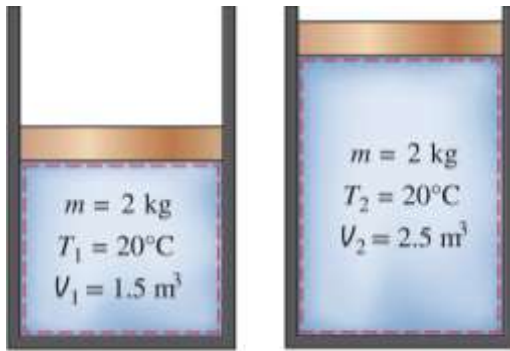
$$v = \frac{V}{m} = \frac{1}{\rho}$$



The density of most gases is proportional to pressure and inversely proportional to temperature.

Liquids and solids, on the other hand, are essentially incompressible substances, and the variation of their density with pressure is usually negligible.

State and Equilibrium (1)



(a) State 1

(b) State 2

A static configuration of a system is called a **state**, if it is completely described by a set of fixed properties.

Thermodynamics deals with **equilibrium states**, i.e. states featuring no unbalanced potentials (or driving forces) within the system.

- **Thermal equilibrium:** If the temperature is the same throughout the entire system.
- **Mechanical equilibrium:** If there is no change in pressure at any point of the system with time.
- **Phase equilibrium:** If a system involves two phases (e.g. vapour and liquid) and when the mass of each phase reaches an equilibrium level and stays there.
- **Chemical equilibrium:** If the chemical composition of a system does not change with time, that is, no chemical reactions occur.



A system is not in thermodynamic equilibrium unless the conditions of all the relevant types of equilibrium are satisfied simultaneously.

State and Equilibrium (2)

- **The State Postulate**

- The number of properties required to fix the state of a system is given by the **state postulate**:
- The state of a simple compressible system is completely specified by two independent, intensive properties.
- Also known as the '**two-property rule**'

- **Simple compressible system:** If a system involves no electrical, magnetic, gravitational, motion, and surface tension effects.
 - If they do exist an additional property for each effect, additional effect is required. (Two-property rule only applies to a simple compressible system)

- **Independent properties:** can be varied independently of each other

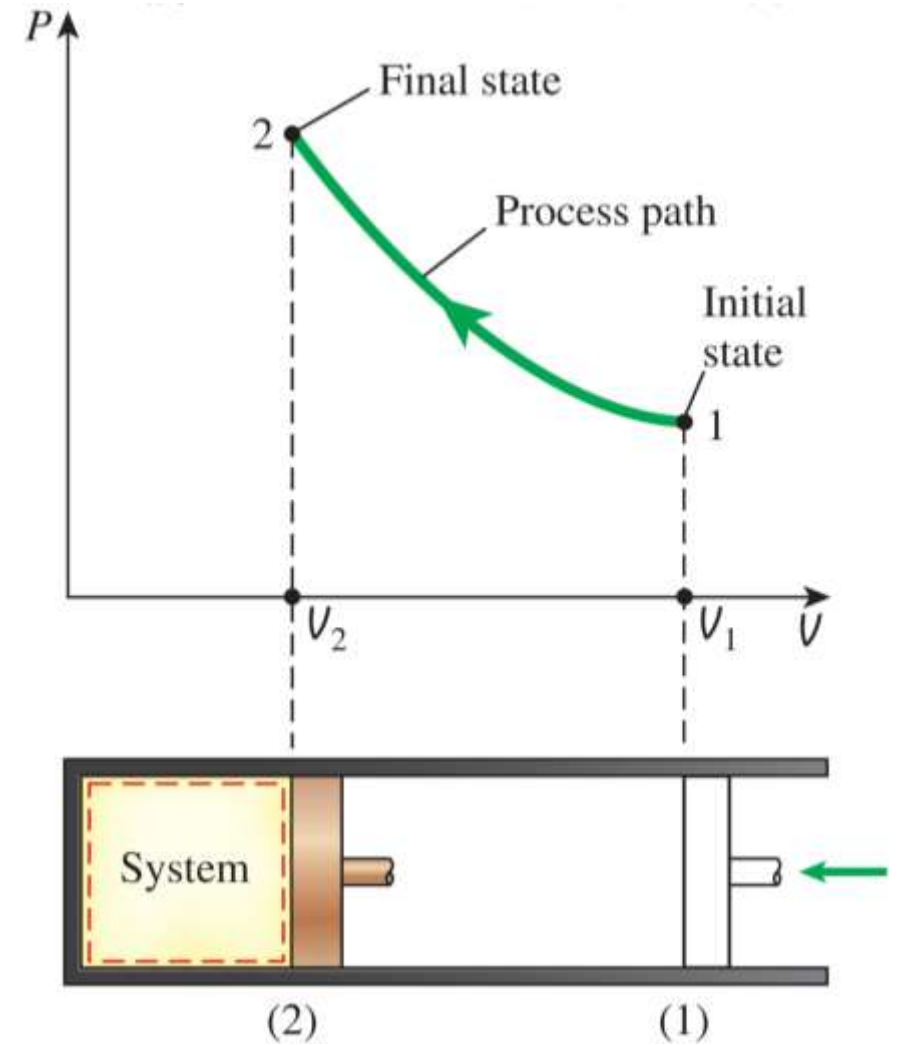
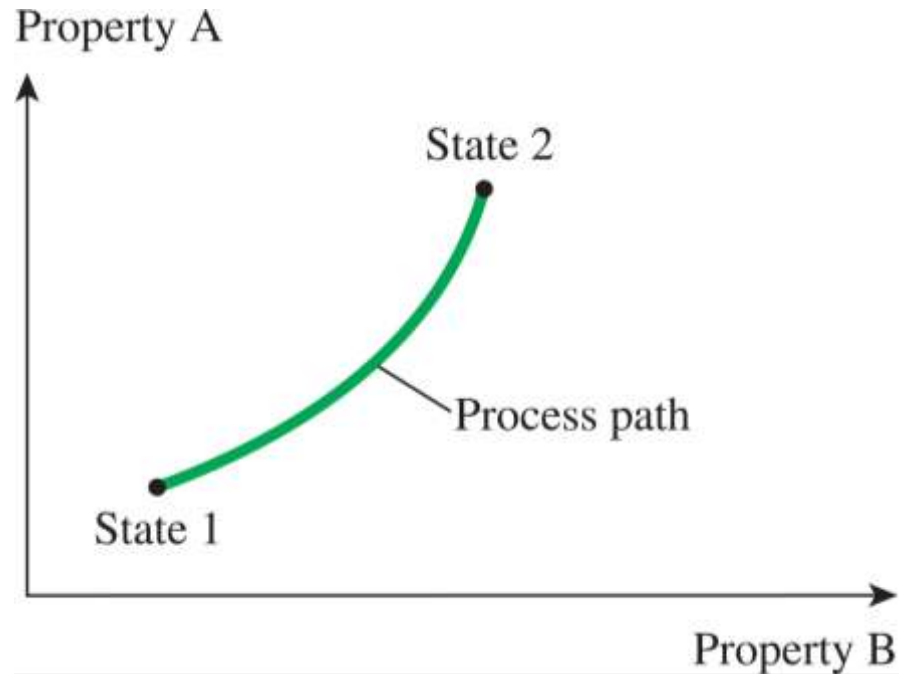


The state of nitrogen is fixed by two independent, intensive properties.

Processes and Cycles (1)

A **process** represents any transformation from one **equilibrium** state (1) to another state (2).

A process is fully described by the initial state (1) and the final state (2) **and** the path from (1) to (2).

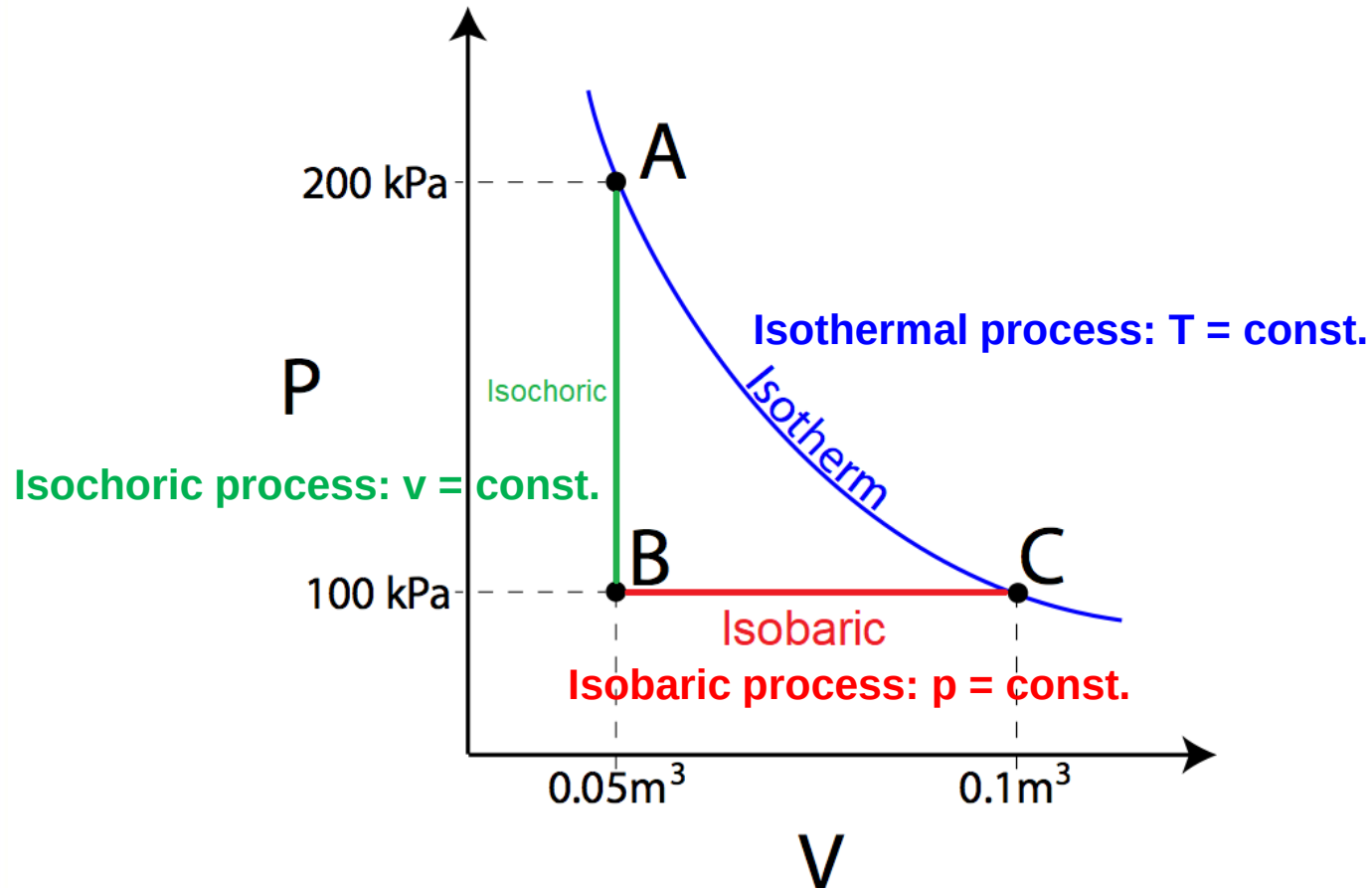


Compression process

Processes and Cycles (2)

A system is said to have undergone a (**thermodynamic**) **cycle** if it returns to its initial state at the end of the process. That is, for a cycle the initial and final states are identical.

Properties, which do not change during a process, are indicated by the prefix “iso”:

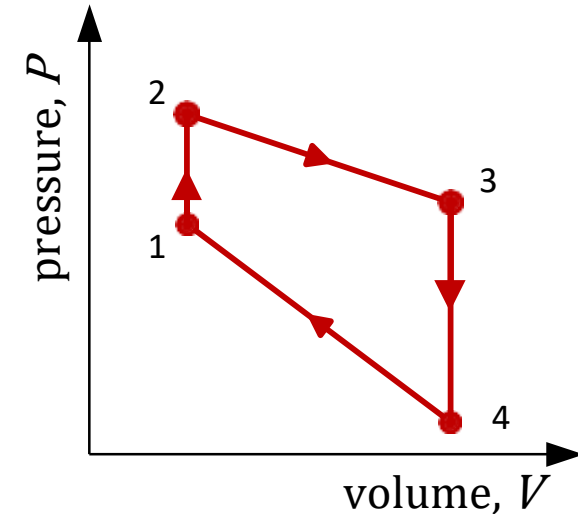


- **Isothermal process:** A process during which the temperature T remains constant.
- **Isobaric process:** A process during which the pressure P remains constant.
- **Isochoric process:** A process during which the volume V or specific volume v remains constant.

Processes and Cycles (3)

- **Cycle:** a series of processes where the initial state and final state are identical.
- For example, for the cycle shown:

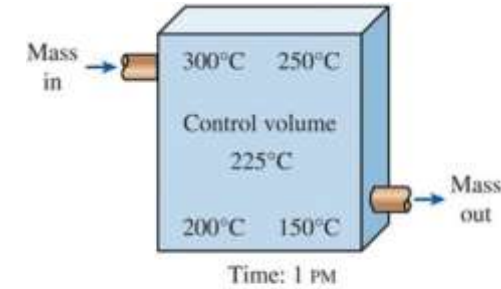
Process	Description
$1 \rightarrow 2$	Constant volume process, pressure increasing
$2 \rightarrow 3$	Expansion process, volume increasing, pressure decreasing,
$3 \rightarrow 4$	Constant volume process, pressure decreasing
$4 \rightarrow 1$	Compression process, volume decreasing, pressure increasing



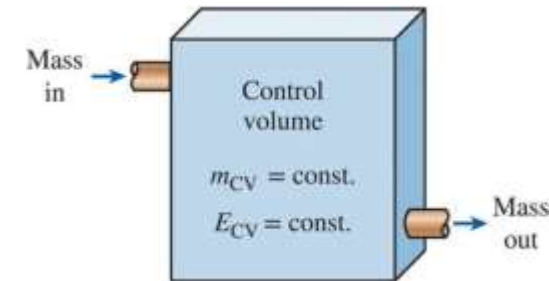
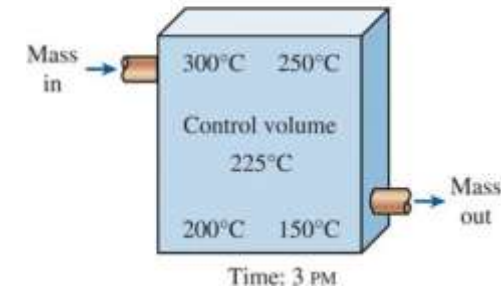
A cycle consisting of four processes plotted on a pressure versus volume ($P - V$) diagram

Processes and Cycles (4)

- **The Steady Flow Process:** A process during which a fluid flows through a control volume steadily.
 - Steady implies *no change with time* (the opposite is unsteady or transient).
 - Many engineering devices operate for long periods of time under steady conditions.
 - These devices (e.g. compressors, turbines, pumps, heat exchangers etc) can be analysed as steady flow devices.
- Under steady flow conditions the mass and energy contents of a control volume (system) remain constant.



During a steady-flow process, fluid properties within the control volume *may change with position but not with time.*



Mass and energy contents do not change under steady flow conditions

Temperature (1)

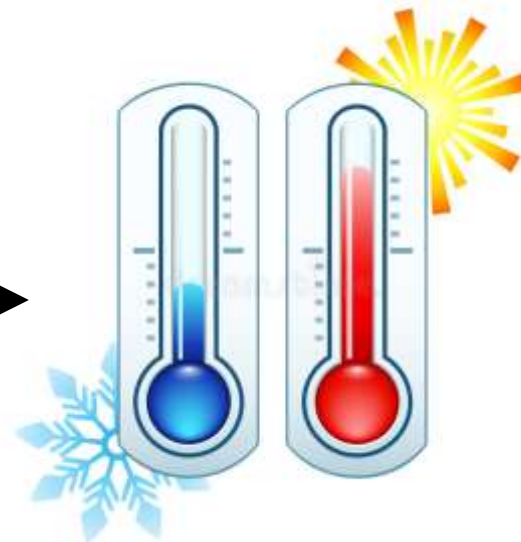
Temperature (T) is a measure of “hotness” or “coldness”

- intensive property
- it is not easy to give an exact definition for it
- we express the level of temperature qualitatively with words like *freezing cold*, *cold*, *warm* and *hot*



However, we cannot assign numerical values to temperatures based on our sensations alone (and senses may be misleading)

Need for temperature scales



Temperature (2)

- **Temperature Scales**

- All temperature scales are based on some easily reproducible states such as the freezing and boiling points of water: *the ice point* and the *steam point*.
- **Ice point:** A mixture of ice and water that is in equilibrium with air saturated with vapor at 1 atm pressure (0 °C or 32 °F).
- **Steam point:** A mixture of liquid water and water vapor (with no air) in equilibrium at 1 atm pressure (100 °C or 212 °F).



Convert to Fahrenheit: $[^{\circ}C] = ([^{\circ}F] - 32) \times \frac{5}{9}$

- **Celsius (°C) and Kelvin (K) scales:** Temperature in SI unit system (*international system of unit*)
- **Fahrenheit (°F) and Rankine (R) scales:** Temperature in English unit system
- **Conversions between scales:**

$$T(K) = T(^{\circ}C) + 273.15$$

(We will use $T(K) = T(^{\circ}C) + 273.15$)

$$T(R) = T(^{\circ}F) + 459.67$$

$$T(R) = 1.8T(K)$$

$$\Delta T(^{\circ}C) = \Delta T(K)$$

$$T(^{\circ}F) = 1.8T(^{\circ}C) + 32$$

Pressure (1)

Pressure (P) is the force applied perpendicular to the surface of an object per unit area over which that force is distributed.

$$P = \frac{F}{A} \left[\frac{N}{m^2} = 1 \text{ pascal (Pa)} \right]$$

- 1 pascal = 1 Pa = 1 N/m²
- 1 bar = 10⁵ N/m²
- 1 Megapascal = 1 MPa = 10⁶ N/m²
- 1 atmosphere = 1 atm = 101325 Pa (absolute pressure)
- (Relative to perfect vacuum)



In thermodynamics we always refer to absolute pressure.

Pressure is always a positive quantity.
Most pressure-measuring devices indicate P_{gauge} .



List of main thermodynamic properties/quantities

Property or quantity	Symbol	Unit	Name on a specific basis	Symbol on a specific basis	Unit on a specific basis	Name on a time rate basis	Symbol on a time rate basis	Unit on a time rate basis
Mass	m	kg	Specific mass (density)	-	-	Mass flow rate	$\dot{m}$	kg/s
Volume	V	m ³	Specific volume	v [= V/m]	m ³ /kg	Volume flow rate	$\dot{V}$	m ³ /s
Energy	E	J	Specific energy	e [= E/m]	J/kg	Energy rate	$\dot{E}$	W [= J/s]
Heat	Q	J	Specific heat	q [= Q/m]	J/kg	Heat transfer rate	$\dot{Q}$	W [= J/s]
Work	W	J	Specific work	w [= W/m]	J/kg	Work rate (Power)	$\dot{W}$	W [= J/s]